Paleomagnetic study on orogenic belt: An example from Early Cretaceous volcanic rocks, Inner Mongolia, China

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Abstract We report paleomagnetic results for Early Cretaceous lava flows collected from the Suhongtu area of Inner Mongolia, the middle part of the Tianshan-Mongolia Fold Belt (TMFB). Rock-magnetic experiments for different lava flows indicate that the main magnetic mineral is pseudo-single-domain (PSD) magnetite. The characteristic high-temperature remanence component is isolated by thermal demagnetization temperature steps between 300℃ and 585℃, which yields a mean direction of $D = 23.6°$, $I = 56.0°$ with $\alpha_{95} = 2.3°$. We interpret this high-temperature remanence component as primary magnetization based mainly upon the petrographic analysis, which shows that the shape of magnetite crystal is relatively rounded square or polygon without internal reflection and deuterogenous phenomenon. The corresponding pole of the high-temperature remanence component is at 71.1°N，200.5°E with $A_{95} = 2.7°$. This Early Cretaceous pole is in good agreement with those for Siberia, North China, and Inner Mongolia, suggesting that these continental blocks had already sutured together in the Early Cretaceous, which would further provide constraints on better understanding of the formation and evolution of the TMFB.

Keywords: orogenic belt, paleomagnetism, Suhongtu of Inner Mongolia, Eurasia.

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Since the 1990s, a large number of paleomagnetic studies have been carried out in the North China block (NCB) and Tarim block[1−8], and more and more geophysicists recently believe that the last collision and convergence between Siberia and the Mongolia-North China plate happened in the Late Jurassic, which was contributed to a paleomagnetic study on these areas by Zhao and his colleagues[2]. However, we lack paleomagnetic results obtained directly from the orogenic belt between Siberia and the Mongolia-North China plate, which may provide us some new evidences on the final timing of convergence. In this paper, with an aim of giving some new constraints on the final timing of convergence between Siberia and the Mongolia-North China plate we present a new Early Cretaceous paleomagnetic result on lava flows collected from the Suhongtu area of Inner Mongolia, the middle part of the TMFB.

1 Geological setting and paleomagnetic sampling

In the middle part of the TMFB, the Early Cretaceous series is well developed in the Suhongtu area of

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Inner Mongolia, which is located at the northeastern area of the Alashan block, about 80 km south to the China-Mongolia boundary. However, the Permian, Triassic and Jurassic systems are absent in this area. Generally, the Early Cretaceous Suhongtu Formation in the Suhongtu area of Inner Mongolia is composed mainly of andesite and basalt with agate, fluorite, and intercalations of grayish green siltstone, silt-limestone, deep-red sandstone and conglomeratic sandstone, succeeding without apparent break the underlying Bayingobi Formation (fig. 1). The Sohongtu Formation yields abundant fossils such as *Eosestheria* sp. and dinosaur *Psittacosaurus*, indicating an Early Cretaceous age[1)].

Paleomagntic samples were collected from Early Cretaceous aubergine andesite-basalt, which is located at the northeast of the Suhongtu agate ore of Inner Mongolia. The bedding attitudes show no dip in the sampling area. According to the thickness of weathering belt, seven periods of volcanic activity were identified in the field, each of them including several lava flows and being distinguished by air hole and oxygenation top. The air hole is usually filled with agate. A total of 74 oriented drill-core samples with 7 sites were collected from the bottom of each lava flow with a portable electric-power drill and were oriented by both magnetic and sun compasses. In the laboratory, a total of 82 samples were cut to the length of 2 cm for paleomagnetic and rock magnetic studies.

2 Rock magnetic properties

In order to identify the rock magnetic behavior of the Early Cretaceous Suhongtu lava flow, some rock magnetic experiments including measurements of temperature-dependence of remanence and hysteresis loop as well as the thin section identification were carried out on one sample of each lava flow. The thermomagnetic analysis of basalt sample determined by VFTB Curie balance in the paleomagnetic laboratory of the Institute of Geology and Geophysics，the Chinese Academy of Sciences, indicates that the collected volcanic samples are unblocked by 585 to 600℃ (fig. 2), suggesting the main magnetic mineral of magnetite of the basalt sample. On the other hand, the irreversible remanence during a heating-cooling cycle in an argon gas atmosphere (fig. 2) may indicate a slight change of magnetic mineral during the heating, however, it is not enough to affect demagnetization behavior of the sample because of very low field of about 10 nT in demagnetization oven.

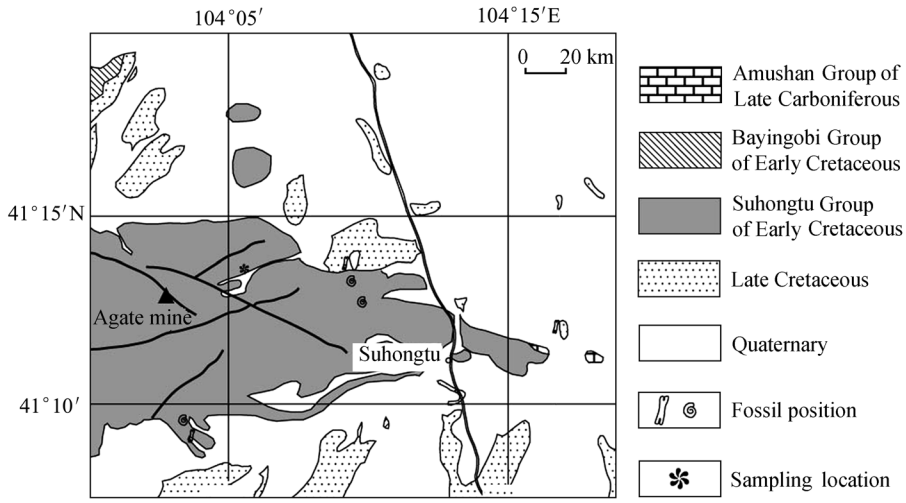


Fig. 1. Sketch map showing the Suhongtu area of Inner Mongolia with an overview of geology and paleomagnetic sampling[1)].

1) Bureau of Geology and Mineral Resources of Ningxia, Regional Geological Report of the PRC, 1:200000 Wuliji Region (K-48- XXVII), 1980.

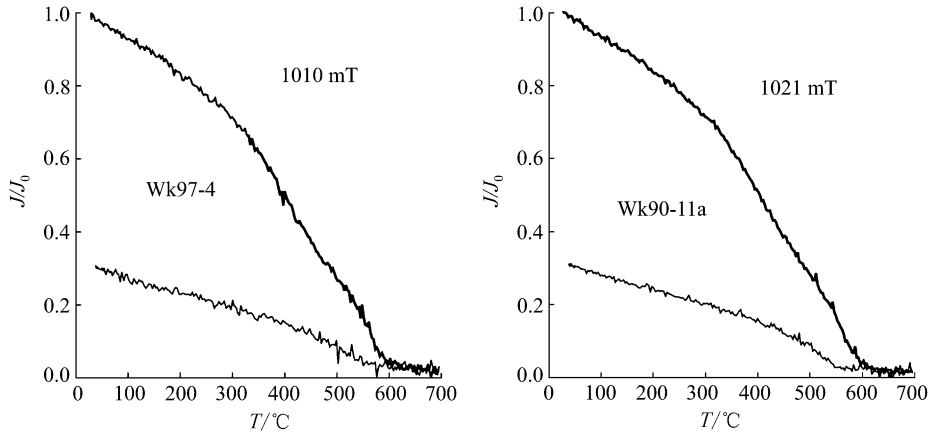


Fig. 2. Typical thermomagnetic behavior of basalt sample. Bold and thin lines indicate the heating and cooling cycle, respectively.

Two typical hysteresis loops, performed on a MicroMag model 2900 alternating gradient magnetometer, are shown in fig. 3. The hysteresis parameters, such as saturation magnetization (M_s), saturation remanence (M_{rs}), coercivity (H_c) and coercivity of remanence (H_{cr}), were calculated after removal of the paramagnetic contributions determined by the slope at high fields. The ratios of hysteresis parameters, such as M_{rs}/M_s and H_{cr}/H_c, as well as the coercivity values, ranging from 25 to 40 mT, indicate that the main magnetic carrier of the remanence to be within pseudo-single domain (PSD) range[9].

In order to further determine the nature of the main magnetic carrier, petrographic analysis was carried out on some pilot samples. Diabasic textures are well developed under a transmission microscope, indicating the main transparent mineral of plagioclases and orthopyroxenes. A little of opaque mineral showing high bright under a reflection light microscope was found in transmission microscope. The shape of crystal is relatively rounded square or polygon without internal reflection and deuterogenous phenomenon (fig. 4), indicating that the main magnetic mineral may be primary magnetite.

In a word, rock magnetic results and petrographic analysis suggest remanence of the Early Cretaceous Suhongtu andesite-basalt resides in primary PSD magnetite.

3 Paleomagnetic results

All the samples were subjected to thermal demagnetization, which was carried out with a MMTD-60 demagnetizer, with 12 temperature steps up to 585℃. Magnetic remanence measurements were performed on a 2G cryogenic magnetometer. Both the demagnetizer and magnetometer are installed within a field free space. Magnetic components for each sample were separated using principal component analysis[10] and calculated using Fisher statistics. Generally, the stepwise thermal demagnetization of the Early Cretaceous basalt samples shows an appearance of a high-temperature component between temperature steps about 300℃ to 585℃ after removal of a low-temperature overprint in the initial stage of demagnetization (fig. 5). Directions of the high-temperature component are nearly paralleled for all the demagnetized samples (table 1). Together with the remanence of these sam-ples residing in primary PSD magnetite, we interpret this high-temperature component as magnetization acquired at, or close to the time of volcano eruption.

The whole section has an *in situ* mean direction of D=23.6°, I=56° with α_{95}=2.3° for the high-temperature component (table 1). This mean high-temperature component yields a corresponding paleomagnetic pole at 200.5°E, 71.1°N with A_{95}=2.7°. The low-temperature component unblocked usually by demagnetization temperature step 300℃ for all the samples has an *in situ* mean direction of D=21.1°, I=58.1° with α_{95}=3.1°, which seems to be similar to direction of the mean high-temperature component of the samples but to that

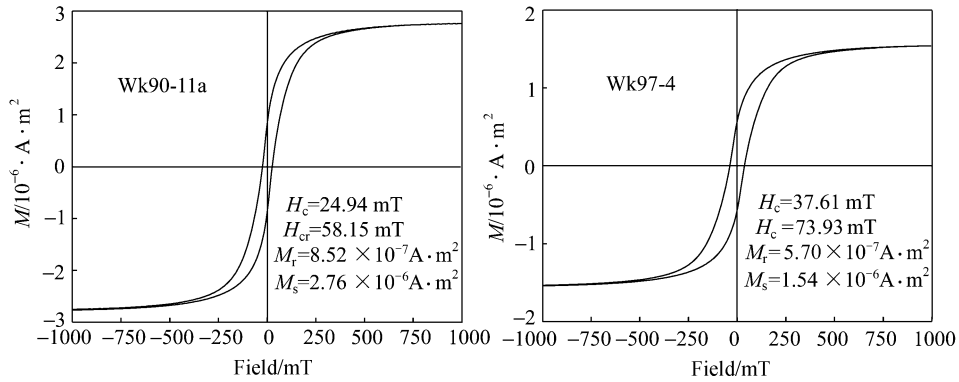


Fig. 3. Hysteresis loops for two pilot basalt samples. Correction of the paramagnetic contributions was carried out by the slope at high fields.

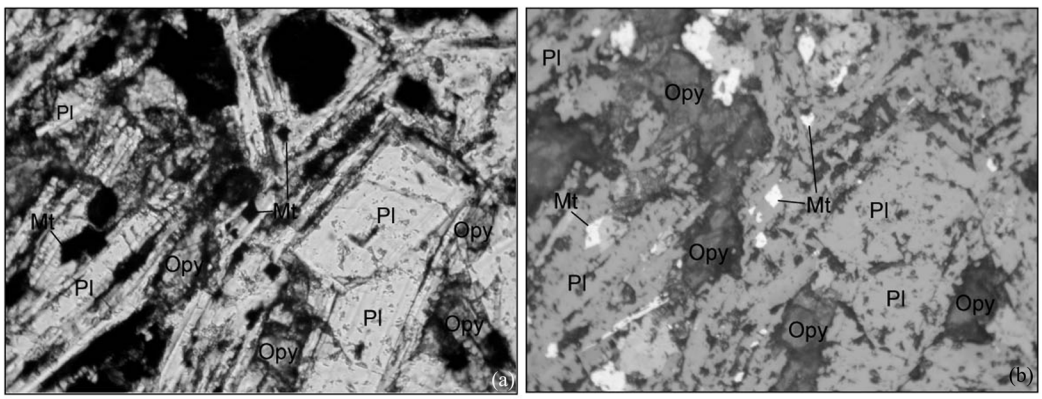


Fig. 4. Rock fabrics under transmission (a) and reflection light (b) microscopes. Pl, Opy, and Mt indicate plagioclase, orthopyroxene, and magnetite, respectively.

of the present Earth field at the sampling site (D= 357.7°, I=60°), indicating a relatively strong magnetic remanence in the sampling area.

It is necessary to check whether geomagnetic secular variation (GSV) is averaged out for igneous rocks or not. The angular standard deviation of the mean virtual geomagnetic pole (VGP), which was calculated from the corresponding VGPs of the site-mean directions of the high-temperature component, is only 3.9°. This angular standard deviation is clearly less than that estimated from the sampling site according to the method of McFadden and his colleagues[11], which gives an angular standard deviation of 15.1°, indicating that the mean VGP obtained from the Early Cretaceous Suhongtu lava flows has averaged out the GSV[12,13], and therefore is available for tectonic discussion.

4 Discussions

Comparison of Early Cretaceous paleopole obtained from the Suhongtu lava flows with coeval ones for the NCB and Eurasian continent (table 2) indicates that these three Early Cretaceous poles are coincident at the 95% confidence level (fig. 6), although the Suhongtu area is located in the TMFB[14]. We note that some Late Mesozoic paleomagnetic studies have been carried out on the middle part of Inner Mongolia and the Ordos Basin of the NCB. These paleomagnetic results suggest that the NCB or NCB-Mongolia plate and the Siberia plate may have already sutured together by the Late Jurassic/Early Cretaceous[2, 15–18]. The consistence between our Early Cretaceous pole from the middle part of the TMFB and coeval ones for the NCB and Siberia further indicates that these tectonic units have completely converged and may have

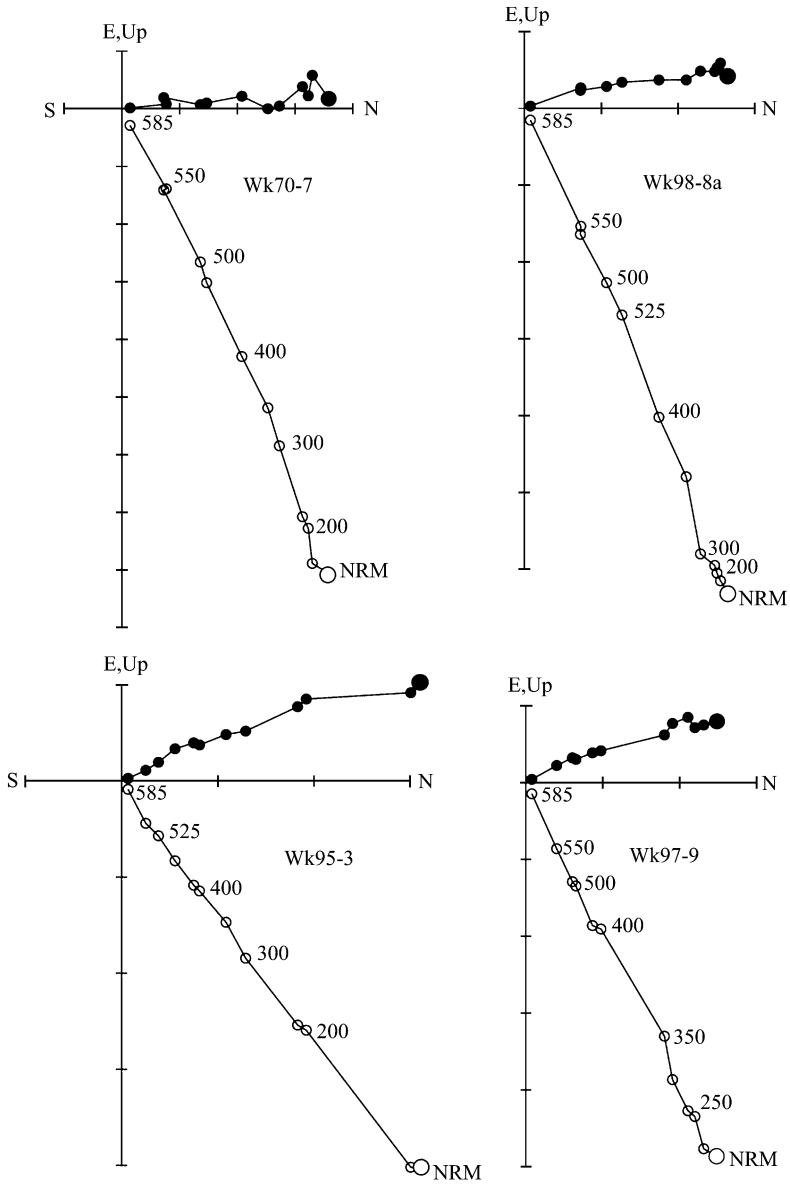


Fig. 5. Orthogonal projections of representative progressive thermal (℃) demagnetization results. The solid (open) symbols represent vector end-points projected onto horizontal (vertical) plane.

Table 1 Summary of the Early Cretaceous paleomagnetic results from the Suhongtu lava flows, Inner Mongolia

Site	*n*/*N*	Low-temperature component			High-temperature component		
		D/(°)	*I*/(°)	α_{95}/(°)	*D*/(°)	*I*/(°)	α_{95}/(°)
Wk1	9/12	25.4	59.1	2.8	26.4	57.6	3.3
Wk2	5/8	25.7	57.5	5.0	24.7	54.2	5.4
Wk3	4/8	27.2	54.8	3.0	28.1	53.1	2.4
Wk4	5/8	10.1	60.0	7.2	19.3	55.2	3.8
Wk5	4/8	21.8	59.1	3.3	25.2	59.2	3.9
Wk6	5/6	22.1	54.7	7.7	24.2	55.2	4.1
Wk7	6/6	13.5	60.2	4.9	16.9	57.1	6.5
Average	7	21.1	58.1	3.1	23.6	56.0	2.3

n/*N*, Number of samples used in the calculation/number of samples demagnetized; *D*/*I*, declination/inclination of direction; α_{95}, radius of 95% confidence limit.

Table 2 List of Early Cretaceous pole positions from the Suhongtu area and its adjacent areas

Blocks	Age	Sampling Location		λ_p/(°)	φ_p/(°)	A_{95}/(°)	Reference
		N/(°)	E/(°)				
Suhongtu	K_1	41.23	104.01	71.1	200.5	2.7	this study
Eurasia	K	—	—	75.2	205.8	5.0	[17]
North China	K_1	35.0	108.0	75.8	208.7	6.8	[18]

λ_p/φ_p, Paleomagnetic pole latitude/longitude; A_{95}, radius of 95% confidence limit.

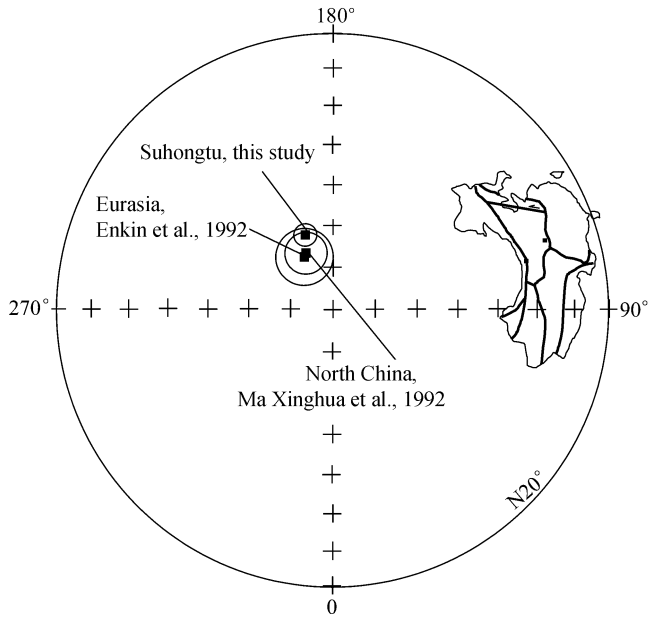


Fig. 6. Equal-area projection of Early Cretaceous paleomagnetic poles with 95% confidence limits for the Suhongtu area, North China block, and Eurasia plate.

formed a united continent in dynamics during the Early Cretaceous time. In addition, it also implies that volcanic rocks from orogenic belt are also available for paleomagnetic study.

On the other hand, comparison of the Early Cretaceous result in this study with a coeval results isolated from North China red beds[19] indicates no or slight difference between these two results, suggesting that the pervasively observed inclination shallowing in Tertiary continental red beds in the Tibetan and central Asian blocks is not the case for Early Cretaceous red beds in North China. The reason of inclination shallowing observed in Tertiary continental red beds from central Asian blocks might result from its special sedimentary and tectonic enviorment. For instance, obvious differences of sedimentary environment could be found between Tertiary and pre-Cenozoic red beds in central Asian blocks, which is obviously caused by the India/Eurasia collision[20]. Consequently, detailed paleomagnetic studies on both Tertiary continental red beds and volcanic rocks in central Asian blocks are obviously useful for discussion of the mechanism of inclination shallowing in that area.

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